Recent Advances in Deep Learning - Spring 2026

Final Examination (Mock Test C - Core Concepts)

Total Marks: 100   Time Allowed: 120 minutes

Part I: True/False Questions (10 questions, 1 mark each, Total: 10 marks)

Please indicate whether the following statements are true or false. Check (✓) for True, and cross (×) for False.

1. The ReLU activation function completely solves the vanishing gradient problem for all inputs. (   )
2. In a Convolutional Neural Network (CNN), max-pooling layers help to introduce a degree of translation invariance to the internal representation. (   )
3. During the inference (testing) phase, Dropout is typically turned off, and the full network is used. (   )
4. Unlike RNNs, the standard Transformer encoder processes all tokens in an input sequence in parallel. (   )
5. L2 Regularization (Weight Decay) penalizes large weights in the network, encouraging the model to use all of its inputs rather than relying heavily on just a few. (   )
6. Batch Normalization normalizes the output of a previous activation layer by subtracting the batch mean and dividing by the batch standard deviation. (   )
7. In Reinforcement Learning, the policy function explicitly maps a state to the maximum expected return. (   )
8. The Adam optimizer calculates an exponential moving average of both the gradient and the squared gradient. (   )
9. An underfitted model will typically have low training error but high validation error. (   )
10. In Self-Attention, the Query (Q) and Key (K) matrices must always have the same dimensionality. (   )

Part II: Multiple Choice Questions (20 questions, 2 marks each, Total: 40 marks)

For each question below, there is only one correct answer. Please select the most appropriate option.

1. Which of the following is the primary reason for using non-linear activation functions in a Multi-Layer Perceptron (MLP)?

    A. To speed up the training process.

    B. To allow the network to learn and represent non-linear decision boundaries.

    C. To prevent the vanishing gradient problem.

    D. To reduce the memory consumption of the model.

2. For a multi-class classification problem with mutually exclusive classes, which output layer activation and loss function combination is standard?

    A. Sigmoid and Mean Squared Error (MSE)

    B. Softmax and Cross-Entropy Loss

    C. ReLU and Cross-Entropy Loss

    D. Tanh and Mean Absolute Error (MAE)

3. What happens if the learning rate in Stochastic Gradient Descent (SGD) is set too high?

    A. The model will suffer from severe overfitting.

    B. The training process will converge very slowly.

    C. The loss may fluctuate drastically and even diverge.

    D. The model will get stuck in a saddle point.

4. Which of the following statements about 1×1 convolutions in CNNs is correct?

    A. They cannot change the number of channels (depth) of the feature map.

    B. They are primarily used to increase the spatial dimensions (width and height).

    C. They act as a channel-wise linear transformation and can be used for dimensionality reduction.

    D. They significantly increase the receptive field of the network.

5. Why is the Long Short-Term Memory (LSTM) network generally preferred over a standard Recurrent Neural Network (RNN)?

    A. LSTM has fewer parameters and trains faster.

    B. LSTM eliminates the need for backpropagation through time (BPTT).

    C. LSTM introduces a cell state and gating mechanisms that mitigate the vanishing gradient problem for long sequences.

    D. LSTM can process input sequences in parallel, unlike standard RNNs.

6. In the context of Transformer models, what is the main purpose of Positional Encoding?

    A. To normalize the input features.

B. To inject information about the relative or absolute position of the tokens in the sequence.

C. To prevent the model from looking at future tokens during decoding.

D. To scale the dot product of Query and Key matrices.

7. Data Augmentation is a technique primarily used to:

   A. Decrease the training time.

   B. Increase the size and diversity of the training set to prevent overfitting.

   C. Automatically tune the hyperparameters of the model.

   D. Reduce the number of parameters in the neural network.

8. In Deep Reinforcement Learning, what is the key difference between Value-based methods (like DQN) and Policy Gradient methods?

   A. Value-based methods output continuous actions, while Policy Gradients output discrete actions.

   B. Value-based methods learn the optimal action-value function (Q-function), while Policy Gradients directly optimize the policy $\pi(a|s)$.

   C. Policy Gradient methods do not use neural networks.

   D. Value-based methods are inherently on-policy, while Policy Gradients are off-policy.

9. Which regularization technique involves adding a penalty term proportional to the sum of the absolute values of the weights to the loss function?

   A. L1 Regularization (Lasso)

   B. L2 Regularization (Ridge / Weight Decay)

   C. Dropout

   D. Early Stopping

10. In a Vision Transformer (ViT), how is a 2D image initially processed before being fed into the Transformer encoder?

    A. By passing it through a deep ResNet backbone first.

    B. By flattening the entire image into a single 1D vector.

    C. By dividing it into non-overlapping patches, and linearly projecting each flattened patch.

    D. By applying a global average pooling layer.

11. Which mechanism in the Transformer Decoder ensures that the prediction for position i can depend only on the known outputs at positions less than i?

    A. Cross-Attention

    B. Scaled Dot-Product Attention

    C. Masked Multi-Head Attention

    D. Positional Encoding

12. If a neural network exhibits high bias and low variance, it is likely:

A. Overfitting the training data.
B. Underfitting the training data.
C. Achieving the optimal generalization.
D. Memorizing the noise in the dataset.

13. In the context of Actor-Critic architectures in Reinforcement Learning, what is the role of the Critic?

A. To select the best action based on the state.
B. To estimate the value function and evaluate the action chosen by the Actor.
C. To update the environment's transition dynamics.
D. To explore random actions to prevent local optima.

14. What is the computational complexity of the standard Self-Attention mechanism with respect to the sequence length N?

A. $O(N)$
B. $O(N \log N)$
C. $O(N^2)$
D. $O(N^3)$

15. Transfer Learning is most effective when:

A. The target dataset is very large and the source dataset is very small.
B. The source task and the target task share low-level features (e.g., edges, shapes in images).
C. The source model was trained on numerical data and the target task is text generation.
D. The pre-trained model is completely randomly initialized before fine-tuning.

16. Proximal Policy Optimization (PPO) is a popular Reinforcement Learning algorithm. What is its core mechanism to ensure training stability?

A. Using a replay buffer with prioritized experience replay.
B. Clipping the probability ratio to prevent the new policy from deviating too far from the old policy.
C. Completely replacing the neural network weights after every episode.
D. Discarding the reward signal and relying entirely on curiosity.

17. Which of the following problems does the Residual Connection (Skip Connection) in ResNet primarily aim to solve?

A. Internal Covariate Shift
B. Degradation problem (training error increasing as network gets deeper)
C. Mode Collapse
D. Imbalanced datasets

18. In a CNN, what is the "Receptive Field"?

A. The size of the output feature map.

B. The region in the input space that a particular CNN feature is looking at.
C. The total number of parameters in a convolutional layer.
D. The optimal learning rate for the specific layer.

19. What is Teacher Forcing in the context of training Sequence-to-Sequence models?

A. Using a larger model to generate pseudo-labels for a smaller model.
B. Feeding the ground truth token from the previous time step as input to the decoder, instead of the model's own prediction.
C. Forcing the model to memorize the training data.
D. Training the model on multiple GPUs simultaneously.

20. Which evaluation metric is most appropriate for a highly imbalanced binary classification dataset (e.g., 99% negative, 1% positive)?

A. Accuracy
B. F1-Score (or Precision/Recall)
C. Mean Squared Error
D. Cross-Entropy Loss

Part III: Short Answer & Essay Questions (Open-end, 5 questions, 10 marks each, Total: 50 marks)

1. **Vanishing Gradients and Architecture Solutions (10 marks)**
The vanishing gradient problem makes it notoriously difficult to train very deep neural networks. (1) Explain the mathematical reason behind the vanishing gradient problem when using the standard Sigmoid activation function. (2) How do modern architectures mitigate this issue? Specifically, explain the mechanisms of the ReLU activation function and Residual Connections (ResNet).

2. **Generalization and Regularization (10 marks)**
Deep neural networks are highly expressive and prone to overfitting. (1) Explain the concepts of Overfitting and Underfitting in terms of Bias and Variance. (2) Choose TWO of the following regularization techniques and explain exactly how they help prevent overfitting during training: Dropout, L2 Regularization (Weight Decay), Early Stopping.

3. **The Transformer Core Mechanisms (10 marks)**
The Transformer architecture has revolutionized sequence modeling. (1) Write down the mathematical formula for Scaled Dot-Product Attention. (2) Explain why the "Multi-Head"

mechanism is beneficial compared to a single attention head. (3) Why is the scaling factor $\frac{1}{\sqrt{d_k}}$ necessary in the attention calculation?

4. **Deep Reinforcement Learning in Continuous Control (10 marks)**
   In modern robotics and autonomous systems, Deep Reinforcement Learning (DRL) is frequently applied to continuous control tasks. (1) Briefly explain the difference between On-policy and Off-policy learning. (2) Why are Policy Gradient or Actor-Critic methods (such as PPO or SAC) generally preferred over traditional Value-based methods (like standard DQN) when dealing with continuous action spaces (e.g., controlling the torque of robotic joints)?

5. **Inductive Biases: CNNs vs. Vision Transformers (10 marks)**
   When processing image data, Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) rely on different design philosophies. (1) Describe the two main "Inductive Biases" inherent in CNNs (Locality and Translation Invariance) and how the convolutional operation enforces them. (2) How does the ViT architecture differ in terms of inductive bias? Discuss how this difference impacts the amount of training data ViTs typically require compared to CNNs to achieve state-of-the-art performance.